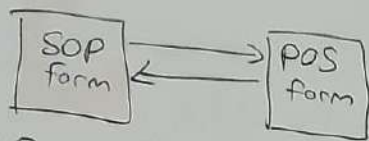


Conversion Between POS and SOP forms



1) From SOP to POS form

$$f(x, y, z) = x \cdot \bar{z} + \bar{x} \cdot y + y \cdot z$$

$$\bar{f} = (f')'$$

$$\bar{f} = \overline{x \cdot \bar{z} + \bar{x} \cdot y + y \cdot z}$$

$$\bar{x} \cdot x \cdot y$$

$$\bar{f} = (\bar{x} + z) \cdot (x + \bar{y}) \cdot (y + \bar{z})$$

$$\bar{f}(x, y, z) = \bar{x} \cdot \bar{y} \cdot \bar{z} + x \cdot y \cdot z$$

$$\bar{f}(x, y, z) = f(x, y, z) = (\bar{x} + \bar{y} + \bar{z}) \cdot (x + y + z)$$

2) From POS to SOP form

$$f(x, y, z) = (x + y) \cdot (\bar{x} + z)$$

$$= \underbrace{x \cdot \bar{x}}_0 + x \cdot z + \bar{x} \cdot y + y \cdot z$$

$$f(x, y, z) = xz + \bar{x}y + yz$$

KARNAUGH MAPS (K-Map)

1) First Order K-Map

Dec	x	f = $\bar{x}$
0	0	1
1	1	0

0	1	0
1	0	1

f

$$f_{\text{sop}} = \bar{x}$$

$$f_{\text{sop}} = \sum m(0)$$

$$f_{\text{pos}} = \bar{x}$$

$$f_{\text{pos}} = \prod M(1)$$

Algebraic

Numeric

2) Second der. k. map

Sinavda bu yontem tercih edilecek

Dec	A	B	f
0	0	0	1
1	0	1	0
2	1	0	0
3	1	1	1

B	0	1
A	1 0	0 1
0	1	0
1	0	1

A	0	1
B	1 0	0 1
0	1	0
1	0	1

colla note

$$f_{sop} = \bar{A} \cdot \bar{B} + A \cdot B$$

$$f_{pos} = (A + \bar{B}) \cdot (\bar{A} + B)$$

$$A \cdot \bar{A} + A \cdot B + \bar{A} \cdot \bar{B} + \bar{B} \cdot B$$

Third Order k. Map

Dec.	A	B	C	f
0	0	0	0	0
1	0	0	1	1
2	0	1	0	1
3	0	1	1	0
4	1	0	0	0
5	1	0	1	1
6	1	1	0	1
7	1	1	1	0

A	BC	00	01	11	10
0	0	0	1	0	1
1	0	0	1	0	1

AB	C	0	1
00	0	1	0
01	1	0	0
11	1	0	0
10	0	1	0

$$f_{sop} = \bar{A} \bar{B} C + \bar{A} B \bar{C} + A \bar{B} C + A B \bar{C}$$

$$f_{sop} = \sum m(1, 2, 5, 6)$$

$$f_{pos} = (A + B + C) \cdot (A + \bar{B} + \bar{C}) \cdot (\bar{A} + B + \bar{C}) \cdot (\bar{A} + \bar{B} + C)$$

$$f_{pos} = \prod M(0, 3, 4, 7)$$



Fourth Order K-Map

Dec	A	B	C	D	f
0	0	0	0	0	0
1	0	0	0	1	1
2	0	0	1	0	1
3	0	0	1	1	0
4	0	1	0	0	0
5	0	1	0	1	0
6	0	1	1	0	1
7	0	1	1	1	1
8	1	0	0	0	1
9	1	0	0	1	0
10	1	0	1	0	0
11	1	0	1	1	0
12	1	1	0	0	1
13	1	1	0	1	1
14	1	1	1	0	0
15	1	1	1	1	1

AB \ CD	00	01	11	10
00	0 ₀	1 ₁	0 ₃	1 ₂
01	0 ₄	0 ₅	1 ₇	1 ₆
11	1 ₁₂	1 ₁₃	1 ₁₅	0 ₁₄
10	1 ₈	0 ₉	0 ₁₁	0 ₁₀

AB \ CD	00	01	11	10
00	0 ₀	1 ₁	0 ₃	1 ₂
01	0 ₄	0 ₅	1 ₇	1 ₆
11	1 ₁₂	1 ₁₃	1 ₁₅	0 ₁₄
10	1 ₈	0 ₉	0 ₁₁	0 ₁₀

K-MAP FUNCTION REDUCTION = MINIMIZATION

Logic Adjacencies $\rightarrow$ Logic Consensus

$$A \underline{\bar{B}C} + \bar{A} \underline{\bar{B}C}$$

$$A \cdot \bar{x} + \bar{A} \cdot x$$

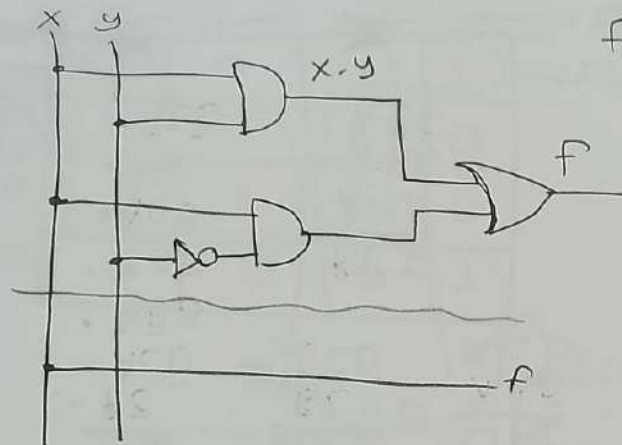
$$\underbrace{(A + \bar{A}) \cdot x}_{1} = 1 \cdot x = x = \bar{B}C \quad \left. \begin{array}{l} 1 \text{ AND} \\ 1 \text{ Inverter} \end{array} \right\} \begin{array}{l} 11 \text{ Input lines} \\ 3 \text{ input lines} \end{array}$$

2 AND
1 OR

2 Inverter

11 Input lines

3 input lines



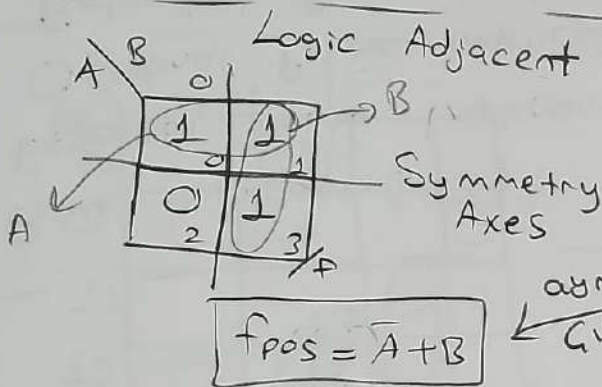
$$F = x \cdot y + x \cdot \bar{y} = x \cdot (y + \bar{y}) = x \cdot 1 = x$$

2 AND
1 OR + 7 input (Lines)
1 NOT

○ Gates + 1 input (line)

Cost Criteria

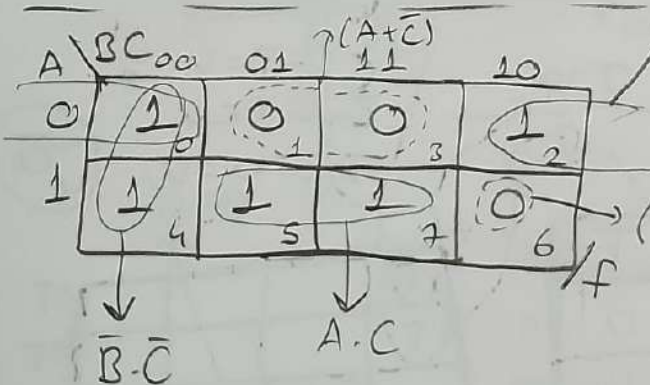
Gates / Inputs



$$f_{sop} = \bar{A} \cdot \bar{B} + \bar{A} \cdot B + A \cdot B =$$

$$(\bar{A} + A) \cdot B = B$$

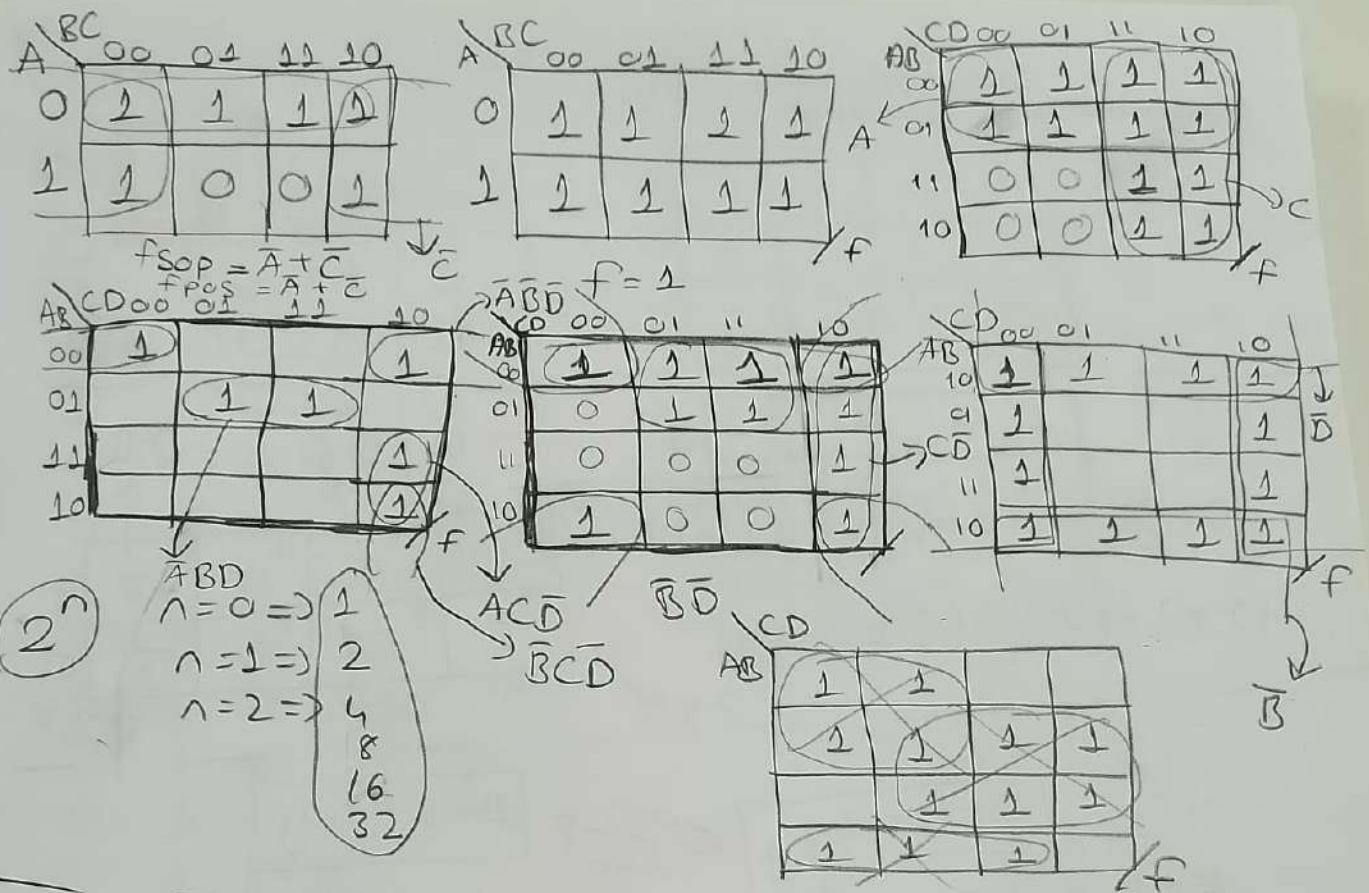
$$f_{sop} = \bar{A} + B$$



$$f_{sop} = (\bar{A} \cdot \bar{C}) + (\bar{B} \cdot \bar{C}) + (A \cdot C)$$

$$f_{pos} = (A + \bar{C}) \cdot (\bar{A} + \bar{B} + C)$$





Prime Implicant (PI)

Essential Prime Implicant (EPI)

Optional Prime Implicant (OPI)

Redundant Prime Implicant (RPI)

<u>PI</u>	<u>EPI</u>	<u>OPI</u>
• $\bar{B}\bar{C}$	$\bar{B}\bar{C}$	$\bar{A}\bar{B}\bar{D}$
• $\bar{A}\bar{C}\bar{D}$	$\bar{B}C\bar{D}$	$\bar{A}CD$
• $\bar{A}\bar{B}\bar{D}$		
• $\bar{A}BD$		
• $\bar{A}CD$		
• $\bar{A}\bar{B}C$		
• $\bar{B}C\bar{D}$		

<u>RPI</u>
$\bar{A}\bar{C}\bar{D}$
$\bar{A}BD$
$\bar{A}\bar{B}C$

The image shows two handwritten Karnaugh maps for a 4-variable logic function. The left map is for variables A, B, C, D and the right map is for variables A, B, C, D.

Left Karnaugh Map (Variables A, B, C, D):

AB \ CD	00	01	11	10
00	1	0	1	1
01	1	1	1	0
11	1	1	0	0
10	0	0	0	1

Groupings and Prime Implicants:

- ACD (Grouping 1s at (00,11), (01,11), (11,11))
- ABD (Grouping 1s at (00,10), (01,10), (11,10))
- ABC (Grouping 1s at (00,01), (01,01), (11,01))
- BCD (Grouping 1s at (01,01), (11,01), (11,11))

Right Karnaugh Map (Variables A, B, C, D):

AB \ CD	00	01	11	10
00	1		1	1
01	1	1	1	
11	1	1		
10				1

Groupings and Prime Implicants:

- ABC (Grouping 1s at (00,01), (01,01), (11,01))
- BCD (Grouping 1s at (01,01), (11,01), (11,11))

$$F = \overbrace{\overline{B}\overline{C} + \overline{B}C\overline{D}}^{EPI} + \overbrace{\overline{A}CD + \overline{A}\overline{B}D}^{OPI}$$

Hand-drawn payoff matrix for a game between AB and CD. The matrix is 4x4 with rows labeled 00, 01, 11, 10 and columns labeled 00, 01, 11, 10. Payoffs are circled in each cell. A large oval encloses the bottom two rows (11, 10) for both players. An arrow labeled f_{pose} points to the right from the bottom right cell.

	CD 00	CD 01	CD 11	CD 10
AB 00	(0, 0)	(0, 0)	(0, 0)	(0, 0)
AB 01	(0, 0)	(0, 0)	(0, 0)	(0, 0)
AB 11	(0, 0)	(0, 0)	(0, 0)	(0, 0)
AB 10	(0, 0)	(0, 0)	(0, 0)	(0, 0)

AB \ CD	00	01	11	10
00			1	
01	1	1	1	
11		1	1	1
10		1		

4. hat der